

DRAFT

Do Not Use Until Posted.

Question #: 1

A 25-year-old man comes to the physician because of a 2-year history of progressively worsening myoclonic epilepsy. Laboratory testing shows elevated blood and CSF lactate levels. He has a family history as follows: his three siblings all have chronic muscle weakness, a maternal uncle with a myopathic disorder; a maternal aunt with progressive dementia who developed ataxia at 37 years; and an 80-year-old maternal grandmother with a life long history of deafness, diabetes, and renal dysfunction. Physical exam shows generalized muscle wasting and weakness, myoclonus, ataxia, and neurosensory hearing loss. He has slow nerve conduction velocities and electroencephalogram (EEG) shows bursts of slow wave and spike complexes. Muscle biopsy identifies abnormal mitochondria, deficient staining for cytochrome oxidase, and ragged-red fibers (muscle fibers with subsarcolemmal mitochondria that stains red with Gomori trichrome stain). Which of the following is the best probability estimate that the children of this patient will inherit the disorder?

- ✓A. 0%
- B. 25%
- C. 33%
- D. 50%
- E. 67%
- F. 75%
- G. 100%

Rationale: This patient has MERFF (myoclonic epilepsy with ragged red fibers) which is mitochondrially inherited, so it typically can only be passed through the maternal line as normally, any mitochondria from the sperm get degraded soon after fertilization. Because of this, children of this man (the patient) are not at risk of inheriting the disorder. Note that there is a high degree of clinical variability in mitochondrial disorders (variable expressivity). This is because of the heteroplasmy.

Note that in late 2018 a report came out which demonstrated paternal transmission of mitochondrial DNA (thank you to all of the students who sent me this article!). Paternal mitochondrial contribution occurs in extraordinary rare situations when the normal paternal mitochondrial DNA degradation system is deficient for some reason. We don't need to concern ourselves with this possibility in most situations.

Question #: 2

A 1-year-old boy is brought to the physician by his parents because of a 3-month history of regression of previously attained developmental milestones. He was sitting when he was 6-months old, however, he now is unable to sit without support. Ophthalmoscopic examination shows a reddish area at the center of macula and a provisional diagnosis of Tay-Sachs disease is made. Samples are obtained for ASO blot analysis for the most common pathogenic variants of the *HEXA* gene. Which of the following is most likely required to perform this technique?

- A. Enzyme linked antibodies
- B. DNA polymerase
- C. SDS polyacrylamide gel electrophoresis
- D. Fluorescently labeled dideoxynucleotides
- ✓E. A labeled single stranded oligonucleotide probe

Rationale: ASO blots use labeled single stranded oligonucleotides complementary to the normal sequence and probes that are complementary to the known mutant alleles of the gene. The probes are allowed to hybridize by complementary base pairing. Presence of a mutant allele is identified as a dark spot.

Enzyme linked antibodies are used in Western blot to identify and detect specific proteins.

DNA polymerase and primers are used for Polymerase Chain reaction (PCR).

SDS-PAGE is used to separate proteins based on size.

Fluorescently labeled dideoxynucleotides are used for Sanger sequencing of DNA.

Question #: 3

A 5-year-old boy is brought to the physician by his mother because of mild intellectual disability, facial anomalies, and a high-pitched nasal voice. He was born with a heart defect, and has had frequent infections. Due to these symptoms, 22q11.2 DS (DiGeorge Syndrome) is suspected. A G-band metaphase karyotype is equivocal, and the direct test of the fragile-X triplet repeat is negative. The most important genetic test for this child would be microarray CGH, but it is not currently offered where this patient is. Which of these other tests would most likely be able to directly test if this patient has 22q11.2 DS?

- A. Sizing of PCR products
- ✓B. Fluorescence in situ hybridization
- C. Southern blot

D. ASO blot

E. cDNA microarray

Rationale: As stated in the vignette, the best test for this patient would be array CGH as it is able to detect copy number variation (genomic deletions and duplications) with relatively high resolution of approximately 30 kb. Importantly, array CGH is able to interrogate the entire genome (meaning it scans the entire genome). Since this vignette states that the array CGH test isn't offered, and there is enough clinical suspicion of 22q11.2 deletion syndrome (DiGeorge syndrome), it makes sense to directly test for this 22q11.2 deletion by FISH. Diagnosis would be established if probes for the critical region of chromosome 22 hybridize to only one of this patient's chromosome 22. Since we are told in the vignette that the patient already had a karyotype, the FISH test can be rapidly done on the same slides that were used for the karyotype.

Sizing of PCR products is important for DNA forensics (simple sequence repeats; SSR) and for diagnosis of triplet repeat expansion disorders. Southern blot makes use of restriction fragment alterations as a result of either mutation of restriction sites, or deletions within a region between restriction sites. Southern blotting is not amenable to automation and requires a high degree of human handling to complete the procedure, so it is typically less expensive than other techniques. In rare cases, such as when PCR fails Southern may be used. ASO blots use oligonucleotide that are specific for a particular allele; ASO blots are useful to detect single (or just a few) base pair changes in a gene. The cDNA microarray is used to detect the expression of many genes on a single chip.

Question #: 4

A 3-month-old girl is brought to the physician by her parents because of congenital malformations including microcephaly, hypertelorism (wide spaced eyes), micrognathia (small jaw), and low set and posteriorly rotated ears. At birth, she was noted to have hypotonia and a characteristic cat-like cry. Genomic testing shows that she has a 2.5 Mb deletion of chromosome 5p and she is diagnosed with Cri du Chat syndrome. Which of the following genetic tests was most likely used to obtain this diagnosis?

A. G-band karyotype

B. Spectral karyotype

C. cDNA microarray

D. Southern blot

✓E. Microarray CGH

F. PCR RFLP

G. ASO Blot

Rationale: The microarray CGH technique is able to detect CNV (copy number variations; i.e. deletions and duplications) that are in the range that this patient has (2.5 Mb). The resolution limit of the array CGH used in most clinical testing is currently at about 30 kb (30,000 base pairs). You should note that if the

resolution of the array is at 30 kb, and there is a 2.5 Mb deletion, there would be approximately 80 probes on the array which would indicate that the patient is missing a copy on her genome ($2,500,000 / 30,000 = \sim 80$). Standard karyotype (G-banding) is unlikely to reveal this deletion because the deletion is probably too small to affect the banding pattern. Typically, a deletion (or duplication) would have to be greater than 5 Mb (or in some cases, 10 Mb) to be detected. Spectral karyotype would not help here, because the entire condensed metaphase chromosome is simply “painted” a color, so a deletion would not be revealed. The cDNA expression array reveals mRNA expression, not copy number variation. The Southern blot would not be helpful here, as the clinical laboratory didn’t know where to look to search for the deletion. In other words, the Southern blot is unable to interrogate the entire genome, whereas the other options are genomic techniques which can.

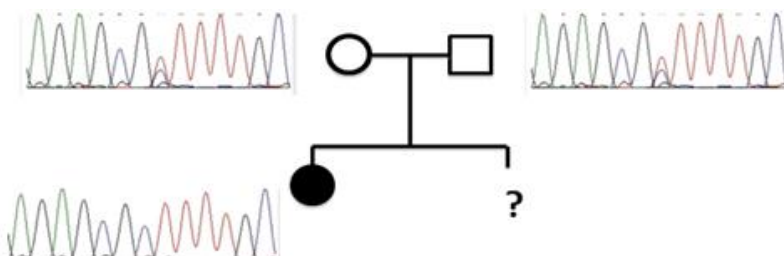
Question #: 5

A 3-year-old girl of consanguineous parents is brought to the physician because of developmental delay, autistic behavior, cleft lip, ptosis, and cupped ears. Proband only exome sequencing shows that that she has a pathogenic mutation that explains her clinical features. Sanger sequencing of the patient and her parents verifies that she has the mutation as shown in the figure. Which of the following would be the best estimate of the probability that these parents will have a second child with the disorder?

- A. 100%
- B. 75%
- C. 67%
- D. 50%
- E. 33%
- ✓F. 25%
- G. 0%

Rationale: From the figure of the DNA sequencing (obtained from Sanger sequencing; chain termination method) we can see that each parent is heterozygous for the seventh base in the chromatograms (count seven peaks from the left). At this position, we can see that there are two different bases detected indicating that each parent is heterozygous. The first child is homozygous at this position, and is affected. Therefore the disorder is autosomal recessive, and there is a 25% chance the next child will be affected.

Attachment:



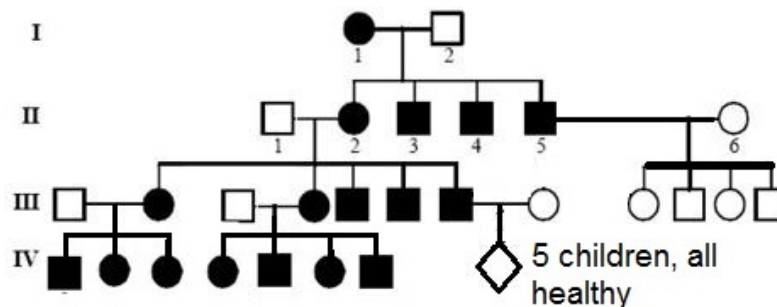
Question #: 6

A 37-year-old woman (individual III-8 of the pedigree) comes to a geneticist to learn more about the disorder on her husband's side of the family. The features of the affected family members vary widely and include muscle twitches, weakness, and progressive stiffness. Some of the affected members display seizures and others display some degree of ataxia. One family member has developed hearing loss, and another loss of visual acuity. A pedigree of the family is shown. Which of the following disorders is likely found in this family?

- A. Becker muscular dystrophy
- ✓ B. Myoclonic epilepsy with ragged red fibers
- C. Prader-Willi syndrome
- D. Marfan syndrome
- E. Lesch-Nyhan syndrome

Rationale: Mitochondrial disorders in a family are inherited maternally and often exhibit pleiotropy (different organ systems affected and mainly high energy requiring organs) and subject to heteroplasmy (percentage of dysfunctional mitochondria in cytoplasm can vary greatly among family members through the maternal line). The pedigree shows all children of an affected female are affected and no male transmission.

Attachment:



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Question #: 7

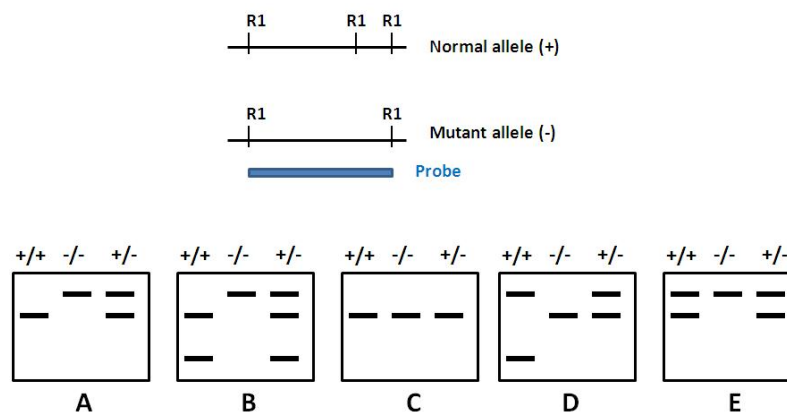
A researcher is performing RFLP analysis on the normal and mutant alleles of a specific gene in the population. The figure shows the location of *EcoRI* restriction endonuclease recognition sites (labeled as R1) for the two alleles (normal and mutant) of the gene. The shaded bar indicates the relative location of a DNA fragment that is used as a probe in the RFLP analysis. DNA samples from three individuals are analyzed: homozygous normal (+/+); homozygous mutant (-/-) and a heterozygous individual (+/-). The DNA from these individuals is digested with

the *EcoRI* enzyme and the RFLP results are shown. Which of the following results would most likely be expected from this analysis?

- A. A
- ✓ B. B
- C. C
- D. D
- E. E

Rationale: Homozygous normal (+/+) - would have two fragments - one short and one larger fragment; Homozygous affected (-/-) - would have only the largest fragment; Heterozygotes (+/-) - would have 3 fragments - the largest from the mutant allele, and two smaller fragments from the normal allele

Attachment:



Question #: 8

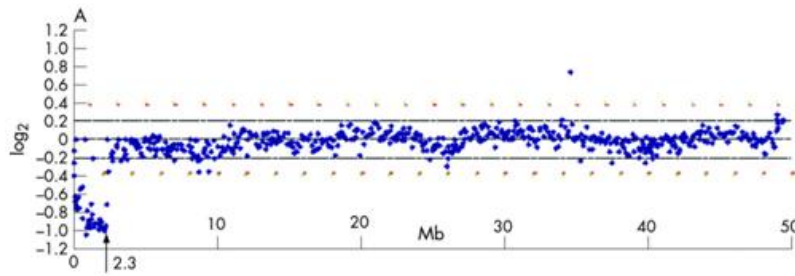
A 13-year-old boy is brought to the physician by his mother because of intellectual impairment and developmental delay. Microarray comparative genome hybridization of his genome shows an alteration on chromosome 4 as shown in the figure. Which of the following is the best interpretation of these data?

- ✓ A. Approximate two megabase deletion on chromosome four
- B. Approximate two megabase insertion on chromosome four
- C. Approximate two megabase duplication on chromosome four
- D. Overexpression of genes on chromosome four
- E. Creation of a dicentric locus on chromosome four

Rationale: The data shows the presence or absence of patient genetic information compared to control. If the amount of patient DNA is equivalent to the amount of control DNA, the "dots" shown in the figure will appear in the area demarked by "0". If the patient has loss of genetic information, the "dots" will be

found in the -1 area. If the patient has more genetic information compared to control, the "dots" will show +2 or more. This case, the patient data shows a loss of genetic information at the tip of the p-arm of chromosome 4 of approximately two megabases.

Attachment:



4190 ExamSoft ID.JPG

Question #: 9

A 50-year-old man comes to the physician for a follow-up examination. Two months earlier he was diagnosed with familial cardiomyopathy. Targeted gene mutation analysis of the α -tropomyosin gene is performed. DNA and RNA are collected and Southern and Northern blot analyses are performed. The data is shown.

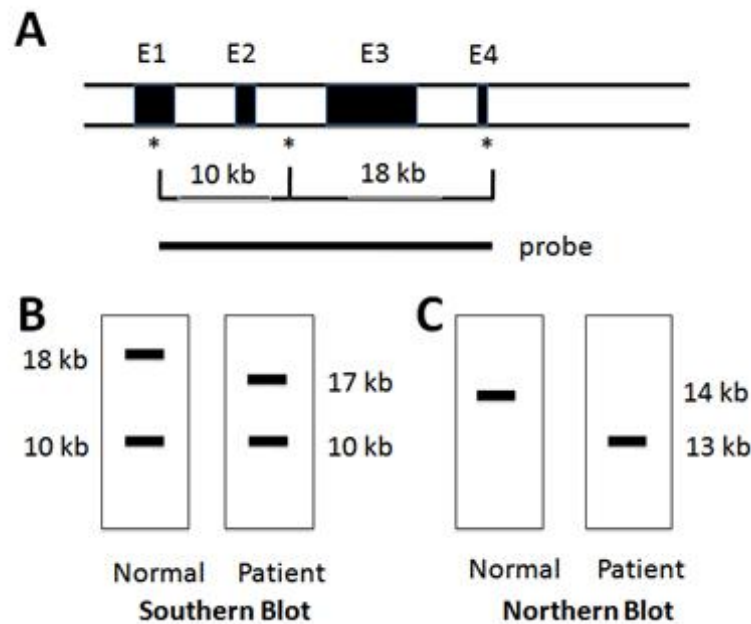
- FIGURE A: *EcoRI* restriction map of the human α -tropomyosin gene. Exons are indicated as shaded boxes, and introns are indicated as open/non-shaded boxes. The location of the *EcoRI* restriction sites are indicated by an asterisk (*), and the distance between these restriction sites is also indicated. The location of the ^{32}P -labeled probed used for the Southern and Northern blot is shown.
- FIGURE B: Southern blot
- FIGURE C: Northern blot

Which of the following is the most likely conclusion from the laboratory data?

- ✓A. There is a deletion of a 1 kb segment of DNA in Exon 3
- B. The sequence of the α -tropomyosin protein is normal
- C. There is a single base mutation that affects a splicing site
- D. The α -tropomyosin gene is not transcribed
- E. There is a deletion of a 1 kb segment of DNA in Intron 3

Rationale: The Southern blot (DNA) shows a shorter fragment which is 17 kb (normal is 18kb); The Northern blot (mRNA) shows a shorter RNA of 13kb (normal 14kb). These results are most likely due to a 1kb deletion in Exon 3 (as it affects the mRNA). A mutation in the splice site would NOT have affected the length of the DNA fragment (as it is a single point mutation). The translated tropomyosin in the patient would be shorter than the normal protein. Deletion in intron 3 would most likely not affect the length of the mRNA, since introns are not present in mature mRNA.

Attachment:



20665.JPG

Question #: 10

A 7-year-old girl is brought to the physician by his parents for a follow-up examination because of developmental delay. Physical examination shows that she is on the 90th percentile for weight and 3rd percentile for height. A picture of the patient is attached. Genetic analysis shows maternal uniparental disomy of chromosome 15. Which of the following syndromes does this patient most likely have?

- A. Angelman syndrome
- B. Rett syndrome
- C. Down syndrome
- ✓D. Prader Willi syndrome
- E. Beckwith Wiedemann syndrome

Rationale: The prevalence of Prader-Willi Syndrome is approx. 1/15,000. Major characteristics is hypotonia, hyperphagia (uncontrollable appetite) global developmental delay, hypogonadism. Facial features are difficulty depending on ethnic background but can include almond shaped eyes and thin upper lip. Most cases of PWS are due to inherited deletion on the Paternally inherited region of chromosome 15 and about 25 % of cases are due to maternal uniparental disomy of chromosome 15.

Attachment:



Girl with PW.JPG

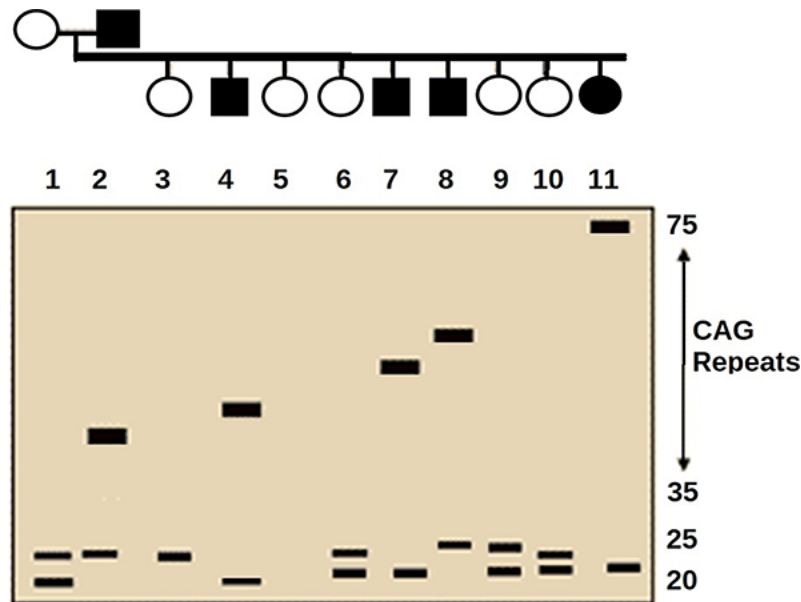
Question #: 11

A 42-year-old woman is brought to the physician by her husband because of a 6-month history of early onset dementia and chorea. Her father also developed similar symptoms when he was in his late sixties. A region within the first exon of a candidate gene was analyzed from all members of the immediate family. The attached figure shows the family pedigree, filled in symbols represent affected individuals, and the gel electrophoresis below each symbol shows the results of the genetic test for each person. Based on these data, which of the following conclusions can most likely be drawn regarding this neurodegenerative disorder?

- A. The disorder is X-linked dominant
- B. The disorder is X-linked recessive
- ✓C. Individual 11 will have the earliest onset age
- D. Definitive diagnosis requires karyotype
- E. Individual 1 (the mother) must be a carrier

Rationale: The triplet expansion repeat undergoes a phenomena of anticipation where the repeat has a chance to be unstable and expand from one generation to the next. Huntington disease is generally more severe if transmitted by the father. In the attached diagram, the proband has a very early age of onset compared to her father and is the individual 11.

Attachment:



huntington.png

Question #: 12

A 2-year-old girl is brought to the pediatric clinic by her parents due to concerns about her developmental progress. They report that she has significant delays in speech and motor skills. She often exhibits a happy demeanor with frequent smiling and laughter. On physical examination, she has a wide mouth, and widely spaced teeth. Her movements are jerky and uncoordinated. Genetic testing is ordered for the patient and her parents, often referred to as "Trio Testing". The results show that the patient has a deletion on the maternal chromosome 15q11-q13. Which of the following is the most likely diagnosis?

- ✓A. Angelman syndrome
- B. Rett syndrome
- C. Down syndrome
- D. Prader Willi syndrome
- E. Fragile X syndrome

Rationale: Angelman syndrome is characterized by severe developmental delays, speech impairment, ataxia, happy demeanor with frequent smiling and laughter. Facial features include a flat occiput, with wide mouth and widely spaced teeth. The most common way to get Angelman is by deletion of the

maternal chromosome 15q11-q13 region that contains the UBE3A gene. Uniparental paternal disomy occurs in about 7% of the cases. Rett syndrome is an X-linked disorder and the affected children are typically female, have developmental delay and have a typical posturing with the hands. Down syndrome most commonly occurs due to non-disjunction of chromosome 21, Prader Willi due to loss of any working copy of the SNRPN gene (either by paternal deletion or maternal uniparental disomy). Fragile X is a triplet repeat disorder and the causative gene is the FMR1 gene. Symptoms of Fragile X are generally milder in females.

Question #: 13

A 4-year-old boy is brought to the physician due to a 1-year history of global developmental delays and facial dysmorphologies. His parents report that he has had multiple congenital anomalies from birth. Physical examination shows a broad forehead, hypertelorism (closely spaced eyes), and a small chin. The physician orders both a SNP and CGH microarray to investigate the underlying genetic cause of the features seen in the patient. Which of the following best describes the utility of using both the SNP and CGH microarray for this patient?

- A. They will detect single nucleotide polymorphisms (SNPs) associated with common diseases
- ✓B. They will detect chromosomal copy number variations (CNVs) and regions of homozygosity
- C. They will give sequence information for the entire genome to identify pathogenic mutations
- D. They will detect both balanced and unbalanced chromosomal rearrangements
- E. They will identify both genomic and mitochondrial DNA mutations

Rationale: CGH microarrays identify chromosomal copy number variations in genomic DNA with much better resolution than karyotype analysis. SNP microarray identifies regions of DNA where there is a loss of heterozygosity which often indicates uniparental disomy, but can also occur in specific tissues in tumor suppressor genes. For a child with global developmental delay and congenital anomalies, physicians often choose to investigate both CNV and uniparental disomy with the same blood sample to cast the largest net in the search for a genetic cause of the clinical features seen in a patient. These microarrays will give no information about single base pair mutations, sequencing, balanced chromosomal rearrangements nor will it give information on Mitochondrial DNA.

Question #: 14

A researcher is investigating the intracellular mechanism of action of a hormone. She plans to develop drugs that mimic the hormone's activity. The hormone facilitates the intracellular release

of sequestered Ca^{2+} from the endoplasmic reticulum. Which of the following messengers most likely performs this action?

- A. Diacylglycerol
- ✓B. Inositol 1, 4, 5-tris-phosphate
- C. Phosphatidylinositol 4,5-bisphosphate
- D. Cyclic-AMP
- E. Cyclic-GMP
- F. Nitric oxide

Rationale: Calcium is released from the endoplasmic reticulum (ER) in the presence of inositol trisphosphate (IP3). IP3 and DAG are formed from the membrane phospholipid, phosphatidylinositol bisphosphate (PIP2) when the alpha-adrenergic receptors are activated by activation of phospholipase C.

Activation of the beta-adrenergic receptors and glucagon receptors increases intracellular cAMP levels. Nitric oxide activates guanylate cyclase which in turn increases cGMP levels within the cells.

Question #: 15

A 62-year-old woman is brought to the emergency department because of a 4-hour history of crushing chest pain that radiates to her left arm. Laboratory studies show elevated serum troponin-T levels. Which of the following is the most likely diagnosis?

- A. Acute pancreatitis
- B. Muscular dystrophy
- ✓C. Myocardial infarction
- D. Acute hepatocellular damage
- E. Obstructive jaundice

Rationale: Increasing troponin-T and CK-MB levels are markers of cardiac injury. Elevated serum amylase and lipase are indicators of acute pancreatitis; CK-MM levels are elevated in muscular dystrophy. Elevated levels of ALT and AST are indicators of Acute hepatocellular damage. Serum ALP levels are elevated in obstructive jaundice.

Question #: 16

A 7-month old boy is brought to the physician by his mother because of a failure to thrive and poor weight gain since birth. Physical examination shows hepatomegaly and jaundice. There is no evidence of biliary obstruction. Which of the following serum marker levels is most likely elevated in this patient?

- A. Amylase
- B. Lipase
- ✓C. Alanine aminotransferase
- D. Alkaline phosphatase
- E. Creatine kinase-MB
- F. Troponin-T

Rationale: Markers for liver disease are ALT and AST. ALP levels are elevated in bile duct obstruction. CK-MB and Troponin-T levels are elevated in myocardial infarction. Serum amylase and lipase are elevated in pancreatic disorders,

Question #: 17

A researcher has identified a drug 'X' which inhibits the G protein-coupled receptor-stimulated phospholipase C activity in intact cell systems. The intracellular changes in signal transduction are observed before and after administration of the drug. Which of the following intracellular changes would most likely be observed following administration of the drug?

- A. Decreased intracellular cAMP levels
- B. Increased intracellular levels of inositol trisphosphate (IP3)
- ✓C. Decreased cytosolic calcium levels
- D. Increased protein kinase C activity
- E. Increased intracellular levels of diacylglycerol
- F. Decreased activity of protein kinase A
- G. Decreased activity of guanylate cyclase

Rationale: Phospholipase C acts on phosphatidylinositol biphosphate (PIP₂) to form DAG and IP₃. IP₃ increases the cytosolic calcium levels. Following administration of a drug which inhibits phospholipase C activity, there would be no increase in IP₃ and DAG and no increase in cytosolic calcium levels.

Question #: 18

A 10-year-old boy is brought to the emergency department by his mother because of a sudden-onset of nausea, vomiting and severe respiratory distress. His mother says that he accidentally spilled a bottle of pesticide (containing organophosphates, DIFP). Ophthalmic examination shows pupillary constriction (miosis). Which of the following is the most likely effect of this toxin on acetylcholinesterase?

- A. Increases the apparent K_m
- ✓B. Binds covalently to the active site
- C. Binds reversibly to a sulfhydryl group
- D. Acts as a feedback inhibitor
- E. Causes reversible inhibition
- F. V_{max} is unchanged

Rationale: Organophosphates including DIFP bind covalently to the active site of the acetylcholinesterase active site (serine residue) and are examples of irreversible inhibitors of the enzyme

Question #: 19

A 67-year-old man comes to the physician for a follow-up examination. During a previous visit 3-months ago, laboratory studies showed his serum LDL cholesterol was 280mg/dL (Normal: <160mg/dL). He was prescribed statins which act as enzyme inhibitors. Which of the following enzyme kinetic changes most likely occurs in the presence of this inhibitor?

- A. The K_m and V_{max} are unchanged
- ✓B. The apparent K_m is increased but V_{max} is the same
- C. The apparent K_m and apparent V_{max} are decreased
- D. The K_m is unchanged but apparent V_{max} is decreased
- E. Statin binds irreversibly to the active site of the enzyme

Rationale: Statins bind reversibly to the active site of HMG CoA reductase enzyme. Statins reduce the binding of substrate to the enzyme and hence increase the apparent K_m . It decreases affinity of the enzyme for its substrate. The V_{max} is not changed.

Non-competitive inhibitors result in lowering of V_{max} and no change in the K_m .

Methotrexate (Inhibits eukaryotic dihydrofolate reductase), trimethoprim (Inhibits bacterial dihydrofolate reductase) are other other examples of competitive inhibitors.

Question #: 20

An investigator is studying the effect of nitric oxide on blood vessels and observes an increase in the intracellular concentration of a specific second messenger in the smooth muscle cells in the walls of the blood vessels. Which of the following enzymes is most likely activated in the smooth muscle cells to produce this second messenger?

- ✓A. Guanylate cyclase
- B. Adenylate cyclase
- C. Phosphodiesterase

- D. Phospholipase C
- E. Protein kinase A
- F. Protein kinase C

Rationale: cGMP is the second messenger produced within the smooth muscle cells when nitric oxide is present. Guanylate cyclase activity is increased in these cells to produce cGMP. Review the actions of the signal transduction proteins listed as the other choices.

Question #: 21

A 30-year-old woman comes to the physician because of a 2-day history of severe watery diarrhea and vomiting. Physical examination shows shrunken skin and dehydration. Stool tests show *Vibrio cholerae*. Which of these modifications is most likely responsible for the pathology in this patient?

- ✓A. ADP ribosylation of the Gs α subunit
- B. ADP ribosylation of the Gi α subunit
- C. ADP ribosylation of the CFTR protein
- D. ADP ribosylation of the EF-2
- E. ADP ribosylation of the Gq α subunit

Rationale: The cholera toxin causes ADP-ribosylation of the Gs-alpha subunit. Phosphorylation of the CFTR protein results in increased efflux of Chloride from intestinal epithelial cells and high stool chloride levels.

Gi-alpha is ADP ribosylated by Pertussis toxin.

Diphtheria toxin causes ADP-ribosylation of Elongation Factor-2 (EF-2) required for translation.

Question #: 22

A 55-year-old man is brought to the emergency room because of a 3-hour history of worsening chest pain radiating down the left arm. The pain began following a game of tennis 3-hours ago. The intensity of the pain has increased progressively and is accompanied by profuse sweating. An ECG is inconclusive, but laboratory tests show myocardial infarction. Which of the following laboratory results best supports the diagnosis?

- A. Elevated serum activities of creatine kinase-BB (CK-BB)
- ✓B. Elevated serum activities of CK-2 and total CK
- C. Increased serum alkaline phosphatase levels
- D. Increased serum amylase and lipase levels

E. Elevated serum alanine aminotransferase levels

Rationale: The markers for myocardial infarction are myoglobin (early but non-specific), troponin I and T (specific), total CK and specifically CK-2 (also known as CK-MB); CK-BB (CK-1) levels are raised in neurological disorders; Serum ALP levels are raised in bone disorders and bile duct obstruction (gall stones in bile duct); Serum ALT levels are raised in patients with liver diseases; Serum amylase and lipase levels are raised in pancreatic disorders (pancreatitis)

Question #: 23

A 16-month-old boy is brought to the physician by his parents for a follow up examination. He has had repeated episodes of respiratory infections in the past two months. Laboratory studies show elevated sweat chloride levels. Genetic tests show that the child is homozygous for the most common mutation in the *CFTR* gene. Which of the following best describes the mutation causing this pathology?

- ✓A. Deletion of a single amino acid in CFTR protein
- B. Deletion of 10 base pairs in the CFTR gene
- C. Deletion of the entire CFTR gene
- D. A glutamate is replaced by valine in the CFTR protein
- E. A frame shift mutation in the CFTR gene
- F. The CFTR protein is shorter by three amino acids

Rationale: The most common mutation of the *CFTR* gene involves a deletion of 3 base pairs (one codon) resulting in deletion of a single amino acid (In-frame mutation).

Question #: 24

A 22-year-old-man comes to the physician for a follow up examination regarding a genetic disorder that was diagnosed a month ago. He has a 3-month history of proximal muscle weakness. Genetic studies show that he has an in-frame deletion in the *DMD* gene which encodes a protein essential for maintaining muscle fiber integrity. Which of the following laboratory findings would most likely be observed in this patient?

- A. Elevated chloride levels in sweat
- ✓B. Reduced staining for dystrophin in muscle biopsy
- C. Absence of dystrophin staining on muscle biopsy
- D. Increased serum creatine kinase-BB levels

E. Anemia and decreased hemoglobin levels

Rationale: The vignette describes a patient with Becker muscle dystrophy. Note the age of presentation (later compared to Duchenne muscular dystrophy). An in-frame deletion most likely results in formation of a truncated protein and as a result, there is reduced staining for dystrophin.

Complete absence of dystrophin staining in muscle biopsy is observed in Duchenne muscle dystrophy which is characterized by a frame-shift mutation resulting in a nonsense mutation and complete absence of dystrophin synthesis.

In BMD and DMD there is increased serum CK-MM levels.

Anemia and low hemoglobin levels are observed in sickle cell anemia or any anemia.

Elevated sweat chloride levels is observed in cystic fibrosis.

Question #: 25

A 3-year-old girl is brought to the physician because of a 6-hour history of intense muscular pain after exercise. She has an older brother who was diagnosed with a genetic disorder. Physical examination shows pallor and generalized pain. A peripheral blood smear is shown. Which of the following molecular changes best explains the clinical presentation in this patient?

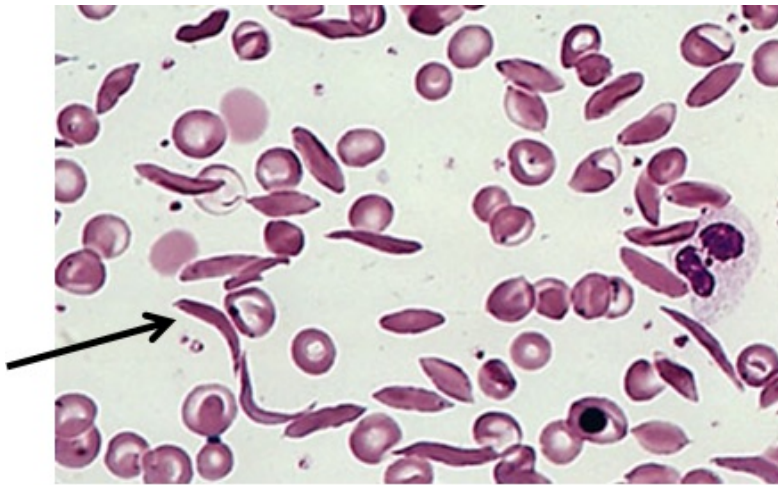
- A. A nonsense mutation in the beta-globin gene
- ✓B. A hydrophobic pocket forms on the surface of Hb
- C. A glutamate to phenylalanine substitution in beta-globin chains
- D. A frameshift mutation in the beta-globin gene
- E. Point mutation in the alpha-globin gene of hemoglobin

Rationale: The vignette describes painful sickling episode in a patient with sickle cell anemia. The peripheral smear shows sickled or crescent shaped RBCs. In HbS, a missense mutation in the beta-globin gene results in the beta globin S protein which has valine (hydrophobic) instead of glutamate (negatively charged acidic amino acid). Presence of Valine creates a hydrophobic sticky patch on the surface of HbS which facilitates polymerization in the deoxy state. HbS is an autosomal recessive disorder.

HbS is a result of a point mutation in the 6th codon of the beta-globin gene. Glu --> Valine which is an example of a missense mutation.

A nonsense mutation results in an amino acid codon --> Stop codon, which results in premature termination of the polypeptide chain.

Attachment:



170033.jpg

Question #: 26

A 49-year-old man comes to the physician because of a 3-month history of loss of appetite, cough and shortness of breath. He has smoked 5-pack year history of cigarette smoking. Physical examination shows hoarseness of voice (dysphonia), emaciation (abnormally thin) and decreased muscle strength. A biopsy specimen obtained from the bronchi via bronchoscopy shows stratified squamous non-keratinized epithelium (abnormal). Following successful treatment, a repeat biopsy shows normal bronchial respiratory epithelium. Which of the following best describes this change?

- A. Dysplasia
- B. Neoplasia
- ✓C. Metaplasia
- D. Hypertrophy
- E. Atrophy

Rationale: Metaplasia involves the replacement of one differentiated cell type by another that is reversible. In this patient we see a change from the abnormal epithelium back to the normal epithelium, which is consistent with metaplasia.

Question #: 27

A 12-year-old boy is brought to the emergency department because of a 1-day history of painful, itchy "sore" on his right leg, nausea, and muscle and joint pain. He received a spider bite at the same site 2 days ago. Physical examination shows a fluid-filled blister that is 2-cm in diameter,

with a red margin. He is diagnosed with a necrotizing ulcer from the venomous bite of a spider. Which of the following best characterizes this type of cell death?

- A. Plasma membrane blebbing
- B. Preservation of homeostasis
- ✓C. Cell swelling
- D. Chromatin aggregation
- E. DNA fragmentation

Rationale: Necrosis is a pathological form of cell death as a result of acute injury (spider bite in this patient) that is characterized by cell swelling and loss of integrity of the plasma membrane, therefore the cell cannot maintain homeostasis and cell leakage results in signs of inflammation (pain and redness in this patient)

Question #: 28

A 45-year-old woman comes to the physician because of a 1-week history of abdominal pain. History shows she frequently drinks alcohol (10 bottles/week). Physical examination shows yellowing of the sclera with shaking of the outstretched hands. A liver biopsy is collected and shows fatty change with accumulation of triglycerides in the hepatocytes' cytoplasm. Which of the following best describes the changes observed in this patient?

- A. Dysplasia
- ✓B. Early cellular injury
- C. Atrophy
- D. Irreversible cellular injury
- E. Neoplasia

Rationale: Fatty change with accumulation of triglycerides in the cytoplasm are reversible, early, morphological changes that may develop following cell injury by a harmful stimulus.

Question #: 29

A 1-month-old boy who was born in a rural clinic is brought to the Pediatric clinic by his parents for review. He was born with a musculoskeletal abnormality of the right foot. Physical examination shows fusion of the right 1st and 2nd toes at the 1st web space. He is diagnosed with Syndactyly (failure of digits to separate into individual appendages due to inadequate

programmed cell death during fetal development). Which of the following best characterizes the defective type of cell death causing this patient's congenital malformation?

- A. Inflammation
- B. Lack of homeostasis
- C. Cell swelling
- ✓D. Chromatin aggregation
- E. Inflammation

Rationale: Apoptosis is physiological process involving programmed cell death that allows for normal formation of some organs (e.g. separation of tissue at distal aspect of the extremities to form individual digits) during fetal development. It is characterized by DNA/chromatin aggregation.

All other answer choices are characteristic of necrosis.

Question #: 30

A 73-year-old woman comes to the physician for a routine examination. She has a history of right breast cancer for which she has completed treatment by radiation therapy 6 weeks ago. Some of the radiation doses were delivered to normal surrounding tissues, resulting in cell injury and the formation of a cystic mass. Tissue collections of the cyst were obtained by surgical resections. Western blot analysis of the cells shows elevated levels of the Bcl-2-associated X (BAX) protein. Which of the following will these cells most likely exhibit?

- ✓A. Release of cytochrome C from mitochondria
- B. Extracellular lactate dehydrogenase
- C. Random DNA degradation
- D. Cell swelling
- E. Positive propidium iodide staining

Rationale: Bcl-2-associated X (BAX) protein is a pro-apoptotic protein that is upregulated during the intrinsic pathway of Apoptosis. This pathway is activated by a death signal (e.g. DNA damage from radiation therapy) with subsequent release of mitochondrial cytochrome C.

All other answer choices are characteristic of Necrosis